

1. What is SQL? What is the difference between SQL and MySQL?

Answer:

SQL stands for Structured Query Language. It is used to communicate with and manage relational databases. We can use SQL to create tables, insert data, retrieve data, update data, and delete data.

MySQL is a relational database management system that uses SQL to work with databases.

So, **SQL is the language, whereas MySQL is a database management system that implements SQL.**

2. What is the difference between DELETE, TRUNCATE, and DROP?

Answer:

DELETE is used to remove specific rows from a table, and we can use a WHERE condition with it.

TRUNCATE removes all rows from a table but keeps the table structure.

DROP removes the entire table, including its structure and data.

For example, if I want to remove selected records, I use DELETE. If I want to remove all records but keep the table, I use TRUNCATE. If I no longer need the table itself, I use DROP.

3. What is a Primary Key? What are its properties?

Answer:

A Primary Key is a column or combination of columns that uniquely identifies each row in a table.

A Primary Key must contain **unique values** and cannot contain **NULL values**.

A table can have only one Primary Key, but that Primary Key can consist of multiple columns, which is called a composite primary key.

4. What is the difference between WHERE and HAVING?

Answer:

Both WHERE and HAVING are used to filter data, but they work at different stages.

WHERE filters individual rows **before grouping**, while HAVING filters groups **after GROUP BY**.

WHERE is generally used for normal row-level conditions, whereas HAVING is commonly used with aggregate functions such as COUNT, SUM, and AVG.

5. What is the difference between CHAR and VARCHAR?

Simple answer:

CHAR is fixed-length, while **VARCHAR is variable-length**.

For example, CHAR(10) reserves space for 10 characters.

VARCHAR(10) can store up to 10 characters and uses space according to the value stored.

So, I use **CHAR for fixed-size data** and **VARCHAR for data whose length can change**.